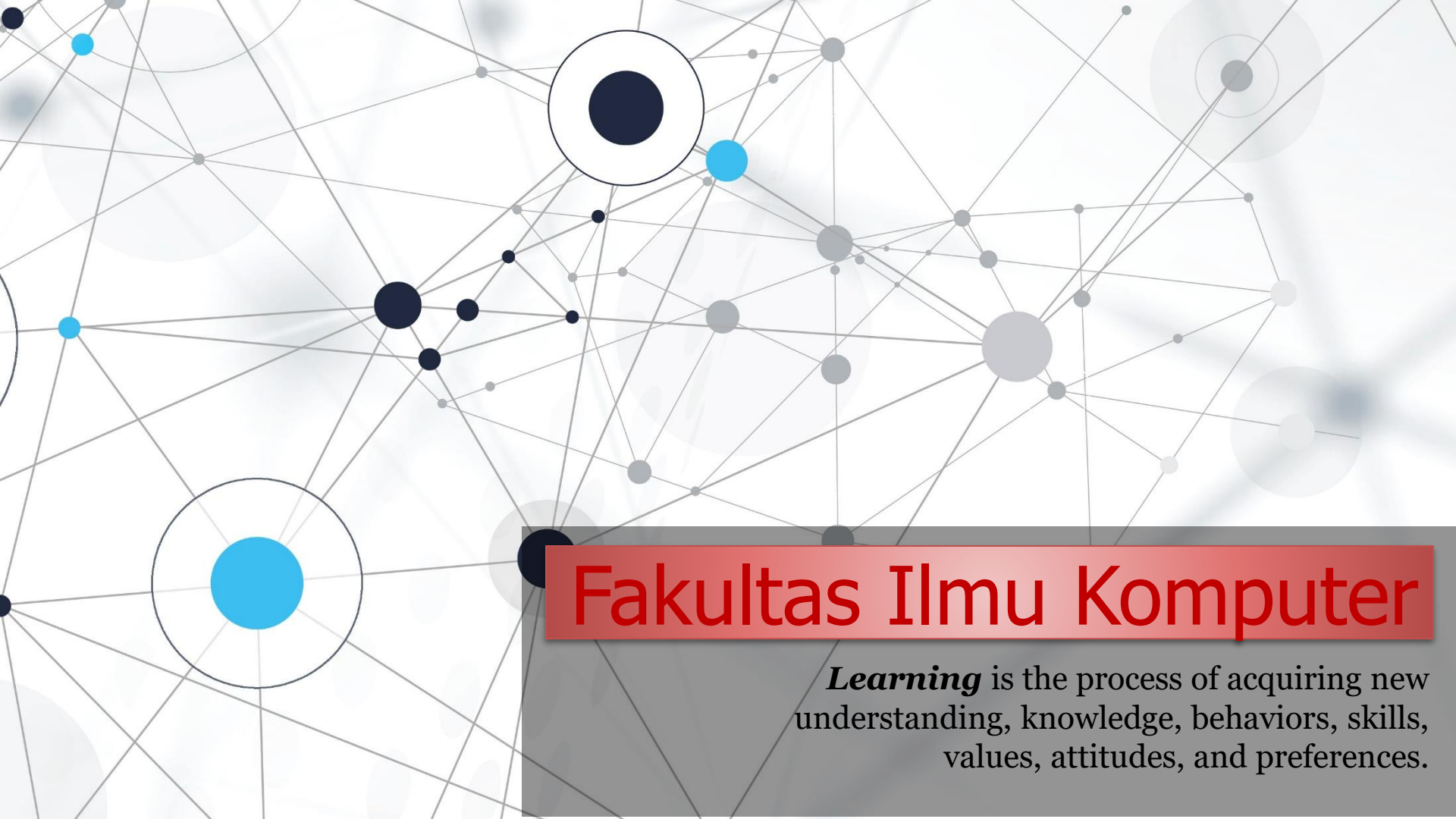




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Magic Constants *in* PHP



PHP Constants (Variables)

PHP Constants

- ❑ **Constants are like variables** except that once they are defined, they cannot be changed or undefined.
- ❑ A constant is an identifier (*name*) for a simple value. The value cannot be changed during the script.
- ❑ A valid constant name starts with a letter or underscore (***no \$ sign before the constant name***).
- ❑ **Constants are Global, constants are automatically global** and can be used across the entire script.
- ❑ **Note:** Unlike variables, ***constants*** are automatically **global** across the entire script.

Create PHP Constant

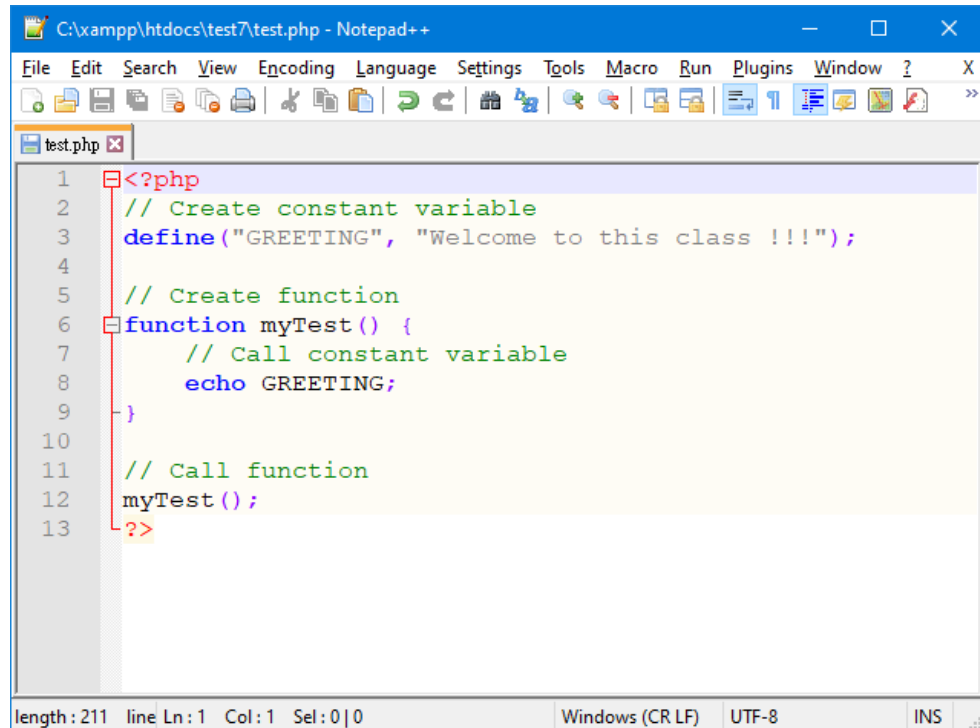
To create a constant, use the **define()** function.

Syntax

```
define(name, value, case-insensitive)
```

Parameters:

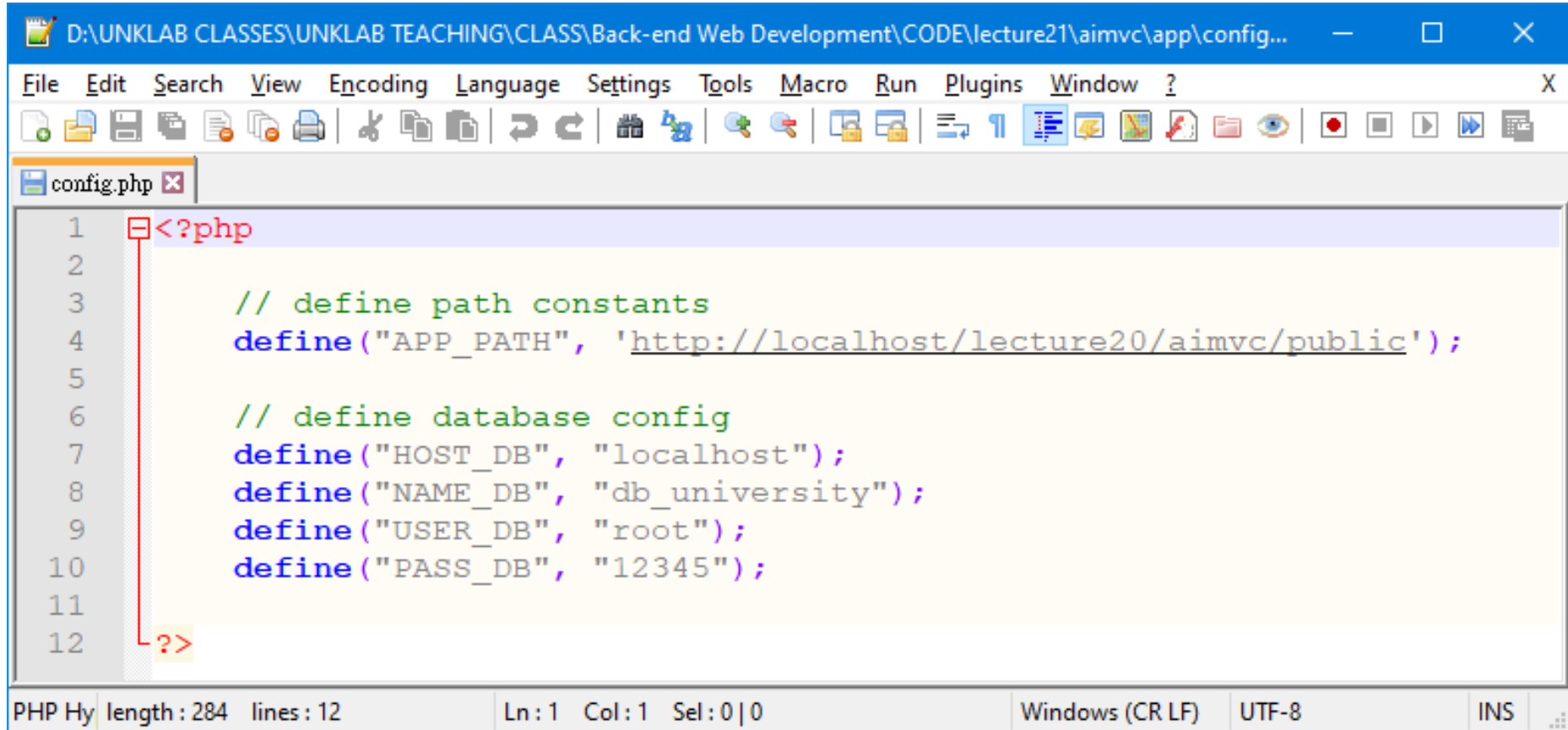
- *name*: Specifies the name of the constant
- *value*: Specifies the value of the constant
- *case-insensitive*: Specifies whether the constant name should be case-insensitive. Default is false



```
C:\xampp\htdocs\test7\test.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? X
test.php
1  <?php
2  // Create constant variable
3  define("GREETING", "Welcome to this class !!!");
4
5  // Create function
6  function myTest() {
7      // Call constant variable
8      echo GREETING;
9  }
10
11 // Call function
12 myTest();
13 ?>
```

length: 211 line Ln: 1 Col: 1 Sel: 0 | 0 Windows (CR LF) UTF-8 INS

Example PHP Constant [Config File]

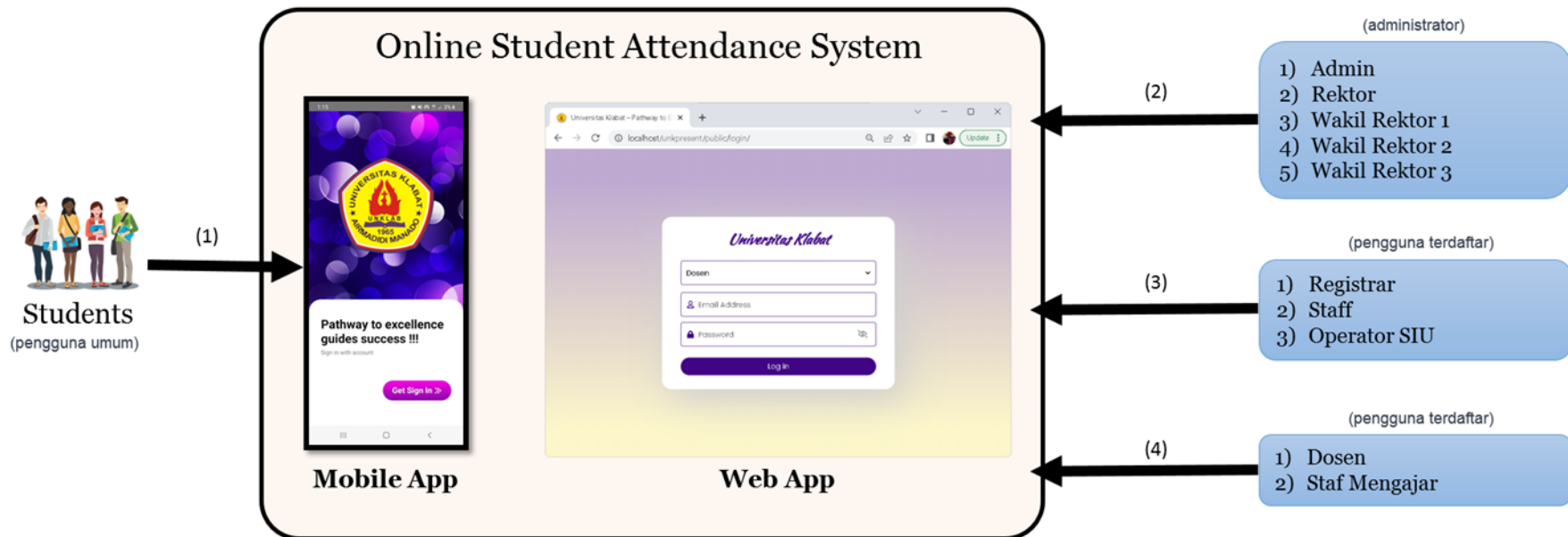


The screenshot shows a code editor window titled "D:\UNKLAB CLASSES\UNKLAB TEACHING\CLASS\Back-end Web Development\CODE\lecture21\aimvc\app\config...". The editor displays a PHP file named "config.php" with the following code:

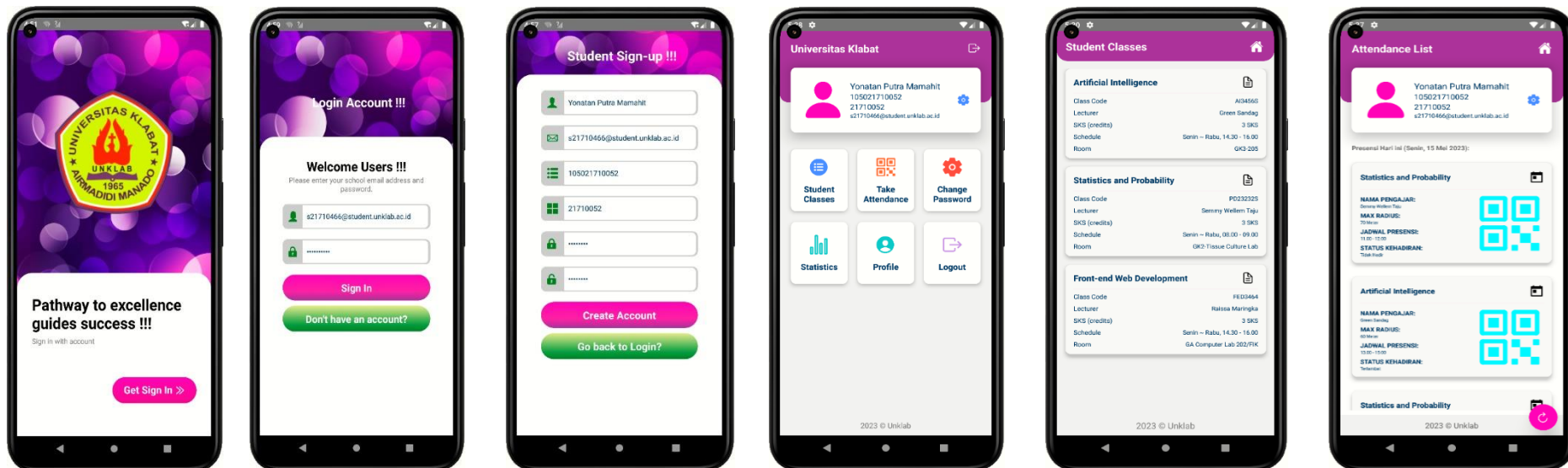
```
1 <?php
2
3 // define path constants
4 define("APP_PATH", 'http://localhost/lecture20/aimvc/public');
5
6 // define database config
7 define("HOST_DB", "localhost");
8 define("NAME_DB", "db_university");
9 define("USER_DB", "root");
10 define("PASS_DB", "12345");
11
12 ?>
```

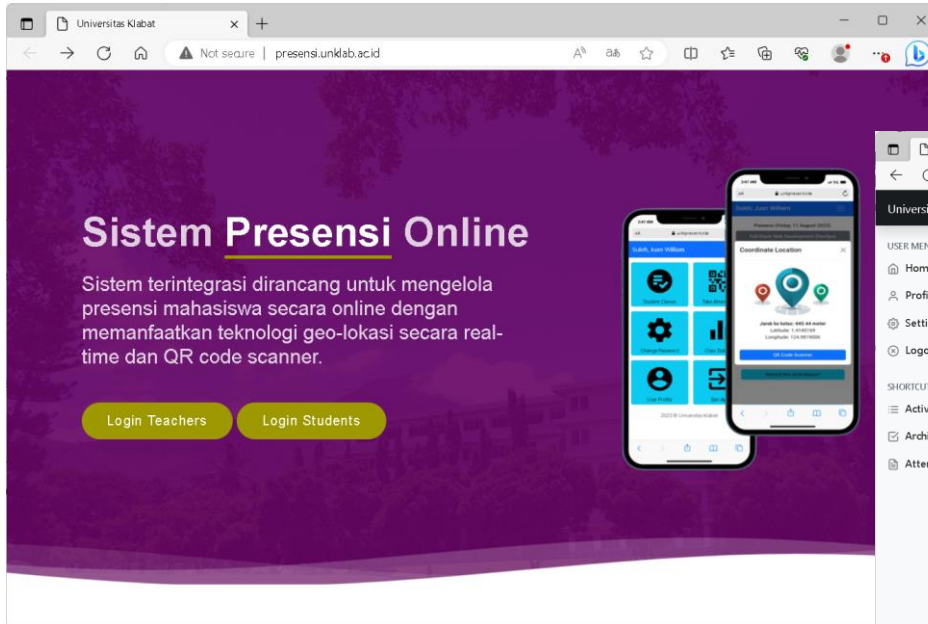
The status bar at the bottom indicates "PHP Hy length : 284 lines : 12", "Ln: 1 Col: 1 Sel: 0 | 0", "Windows (CR LF)", "UTF-8", and "INS".

Menurut kalian, apakah ini aman?

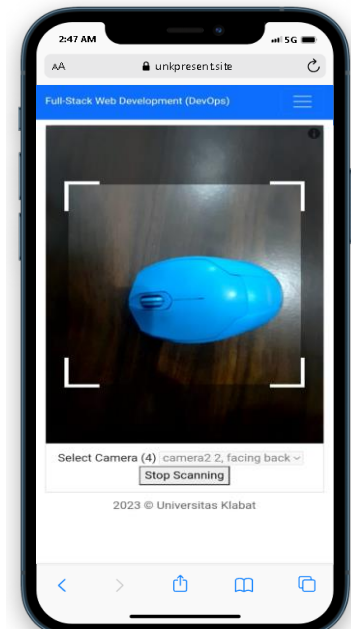
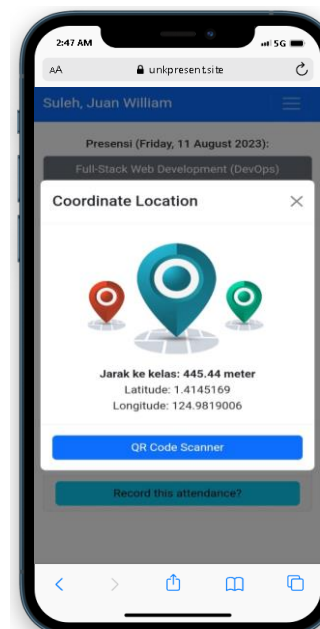
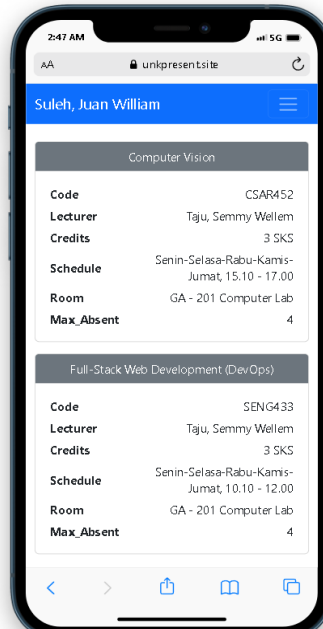
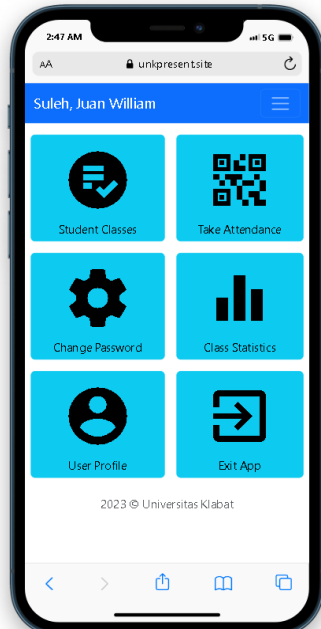
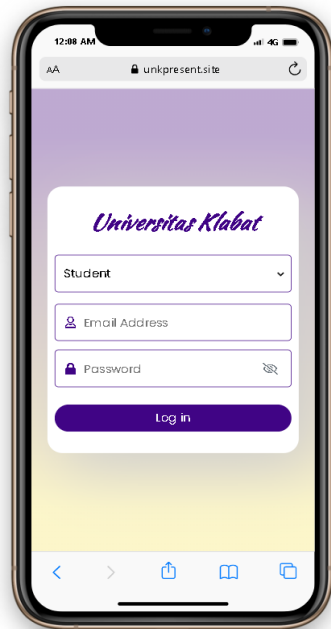


- 1) **Pengguna umum** hanya memiliki akses terbatas ke fitur dan informasi dalam sistem.
- 2) **Pengguna terdaftar** memiliki hak akses lebih seperti mengunggah atau mengedit.
- 3) **Administrator** memiliki hak akses penuh dan dapat mengatur dan mengelola sistem secara keseluruhan.

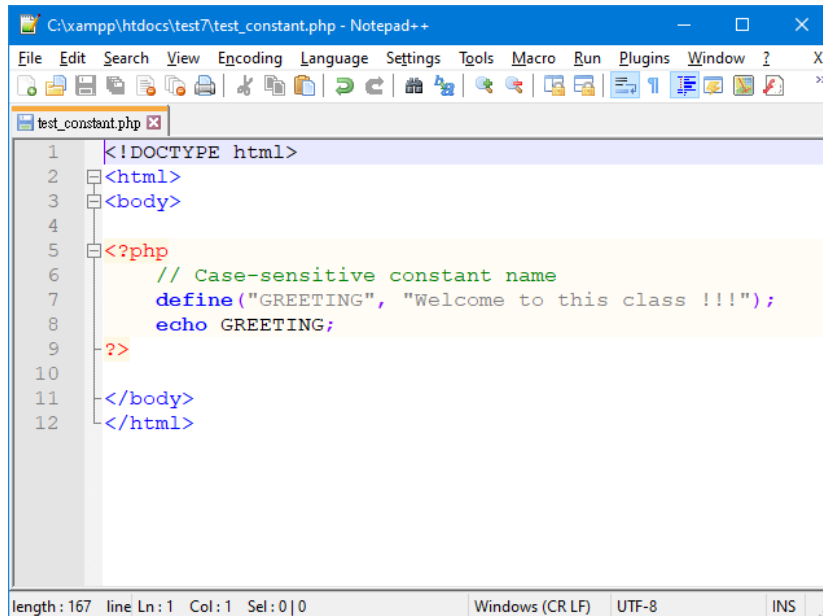




No.	Students	Number meetings?	Hadir	Late	Izin	Alpa
1	Abram, Mogandi Timotius S2200060@student.unklab.ac.id	19 Meetings	19 times	0 times	0 times	0 times
2	Amping, Bryan S2200047@student.unklab.ac.id	19 Meetings	18 times	1 times	0 times	0 times
3	Dumais, Ade Pricilia S2200153@student.unklab.ac.id	19 Meetings	19 times	0 times	0 times	0 times
4	Haerani, Christian Solafide S2200045@student.unklab.ac.id	19 Meetings	14 times	3 times	0 times	2 times
5	Kabo, Deo Timothy S2200637@student.unklab.ac.id	19 Meetings	16 times	2 times	1 times	0 times
6	Korengkeng, Vito Julio S2200432@student.unklab.ac.id	19 Meetings	18 times	1 times	0 times	0 times
7	Mumu, Griffin Meidy S2200054@student.unklab.ac.id	19 Meetings	18 times	0 times	0 times	1 times

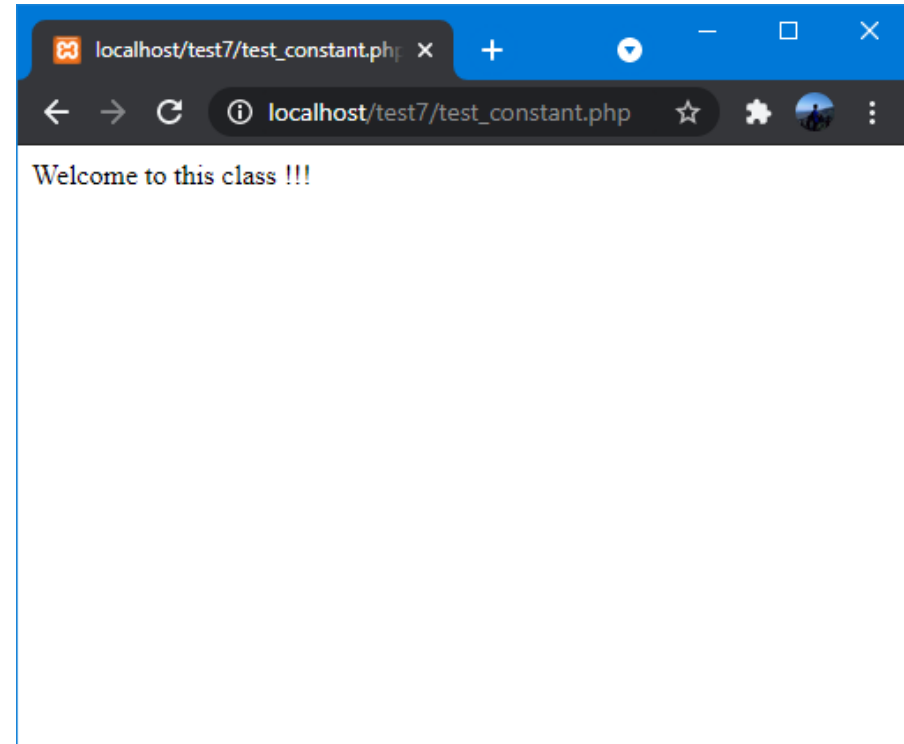


Create Constant *with* Case-sensitive



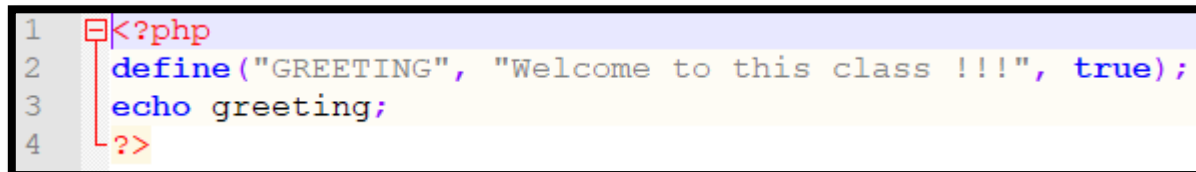
```
1 <!DOCTYPE html>
2 <html>
3 <body>
4
5 <?php
6     // Case-sensitive constant name
7     define("GREETING", "Welcome to this class !!!");
8     echo GREETING;
9 ?>
10
11 </body>
12 </html>
```

length: 167 line Ln: 1 Col: 1 Sel: 0|0 Windows (CR LF) UTF-8 INS



```
localhost/test7/test_constant.php x +
localhost/test7/test_constant.php
Welcome to this class !!!
```

Create a constant with a **case-insensitive** name:



```
1 <?php
2 define("GREETING", "Welcome to this class !!!", true);
3 echo greeting;
4 ?>
```



What is Magic Constants?



Magic constant adalah Predefined (*siapa pakai*) konstanta yang memiliki fungsi khusus atau ajaib. Contohnya, value dari *Magic constant* `__LINE__` tergantung dari *line* dimana *script* tersebut diletakan.



Cara penulisan *magic constant* diawali dengan dari 2 *underscore* atau garis bawah (`__`) dan diakhiri dengan 2 *underscore* (*begin with two underscores and end with two underscores*).

Mengapa penulisan **magic constant** atau predefined (*siapa pakai*) konstanta variables sebaiknya harus diawali dan diakhiri dengan dua underscore?



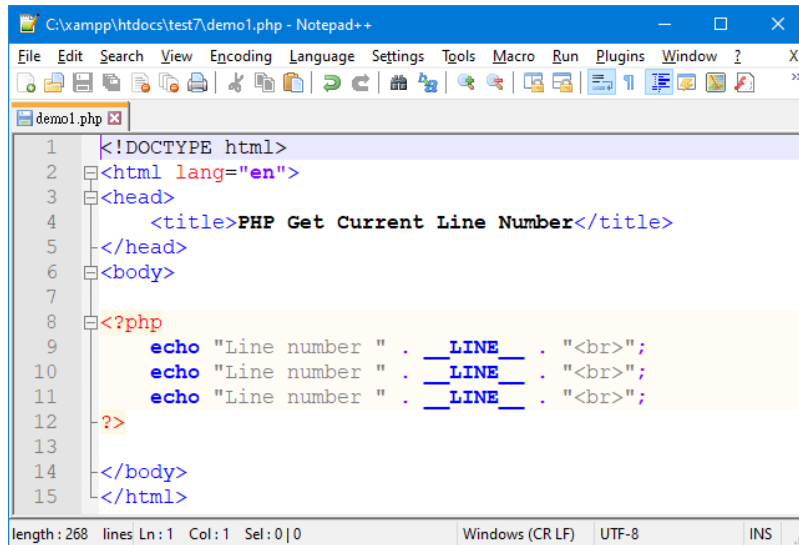
Magic Constants



- ❑ PHP memiliki banyak konstanta yang siap pakai (*predefined*), tanpa harus dibuat.
- ❑ Konstanta adalah sebuah nilai yang tidak akan pernah berubah dan tidak bisa diubah.
- ❑ Namun, ada 9 konstanta ajaib (*magic constant*) di PHP yang akan selalu berubah nilainya tergantung pada penggunaan-nya.

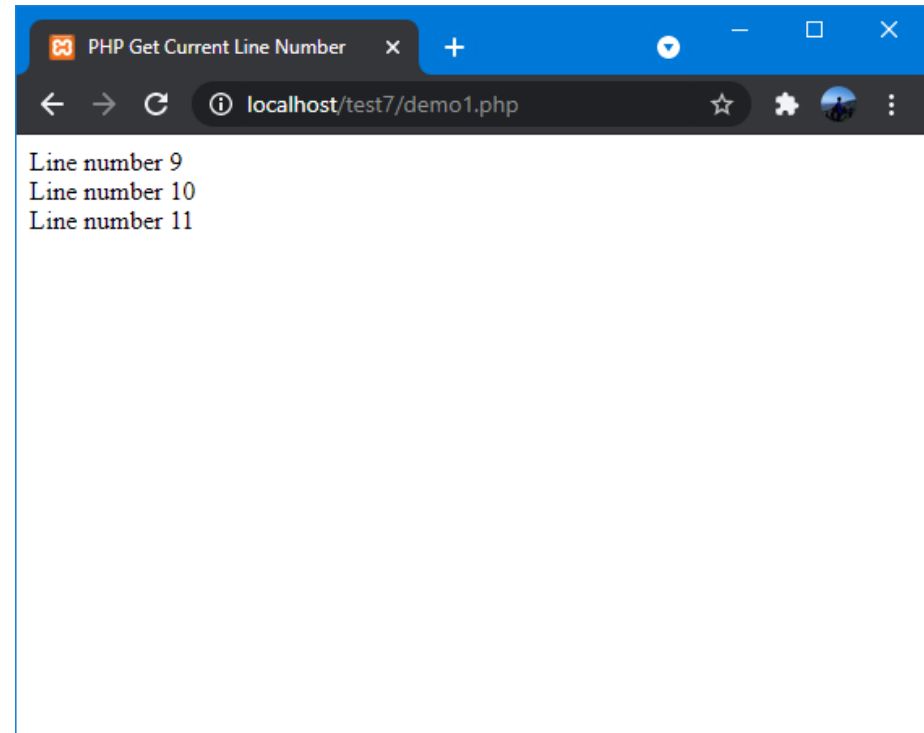
1. Konstanta `__LINE__`

Konstanta `__LINE__` akan berisi sebuah nilai yang menyatakan nomer baris saat itu (*current line number of the file*).



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <title>PHP Get Current Line Number</title>
5 </head>
6 <body>
7
8 <?php
9     echo "Line number " . __LINE__ . "<br>";
10    echo "Line number " . __LINE__ . "<br>";
11    echo "Line number " . __LINE__ . "<br>";
12 >?>
13
14 </body>
15 </html>
```

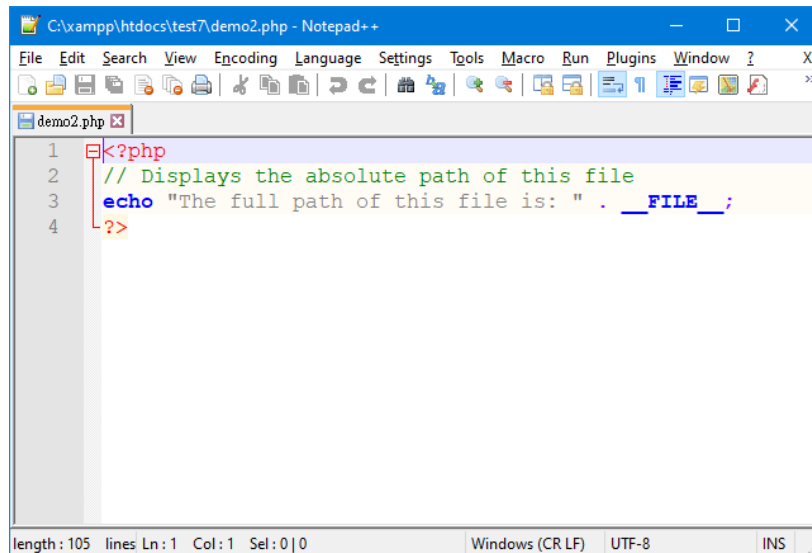
length: 268 lines Ln: 1 Col: 1 Sel: 0 | 0 Windows (CR LF) UTF-8 INS



```
Line number 9
Line number 10
Line number 11
```

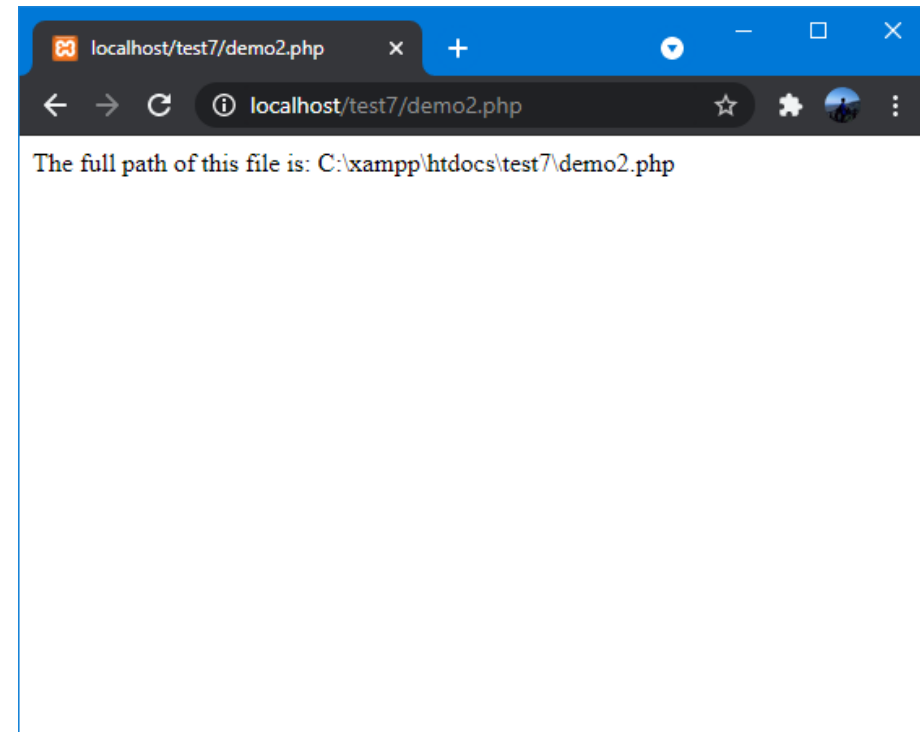
2. Konstanta `__FILE__`

Konstanta `__FILE__` akan berisi alamat lengkap (*path*) dari file PHP (*return full path and name of the PHP file that's being executed*).



```
1 <?php
2 // Displays the absolute path of this file
3 echo "The full path of this file is: " . __FILE__;
4 ?>
```

length: 105 lines: 1 Col: 1 Sel: 0 | 0

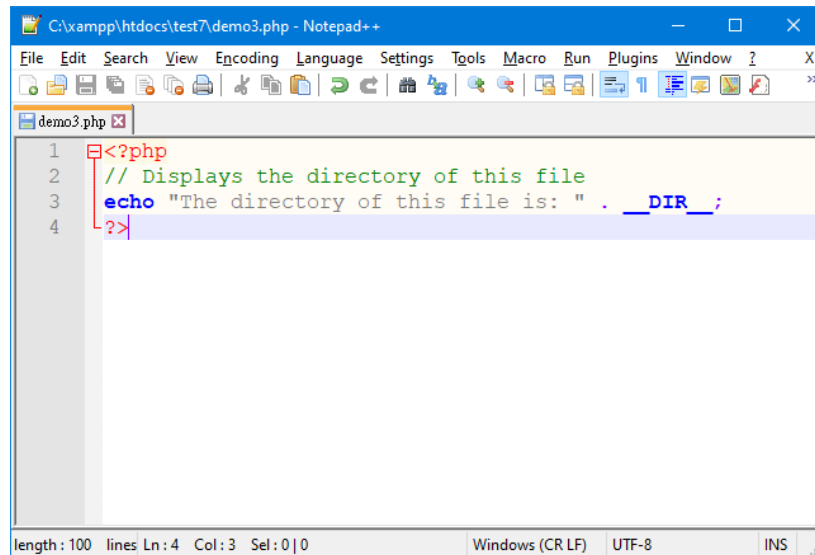


localhost/test7/demo2.php

The full path of this file is: C:\xampp\htdocs\test7\demo2.php

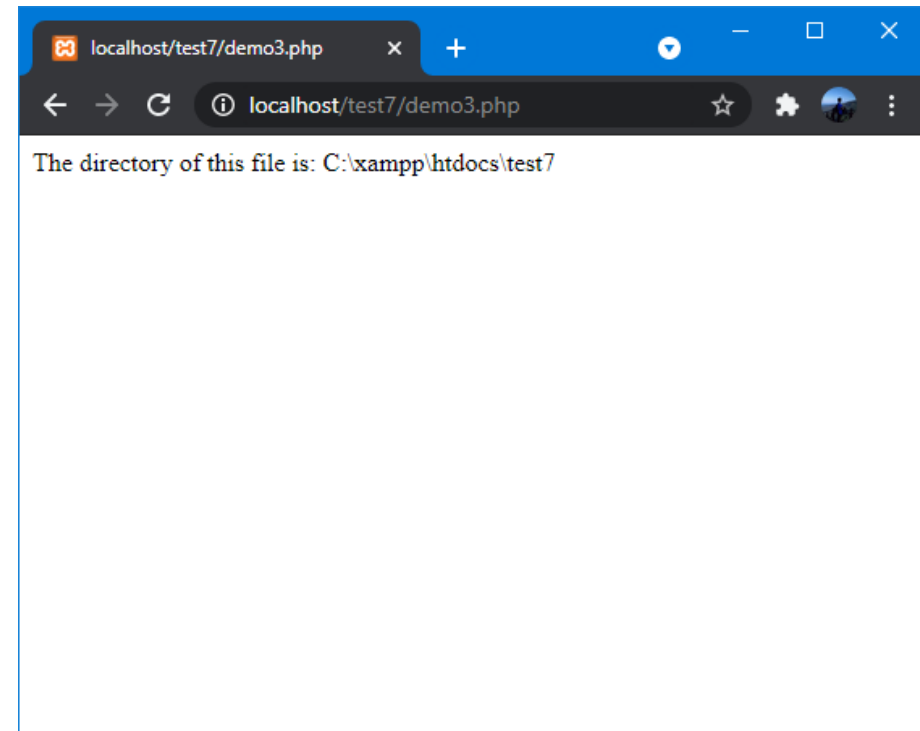
3. Konstanta `__DIR__`

Hampir sama dengan konstanta `__FILE__`, konstanta `__DIR__` akan berisi alamat direktori dari file PHP (*return the directory of the file*).



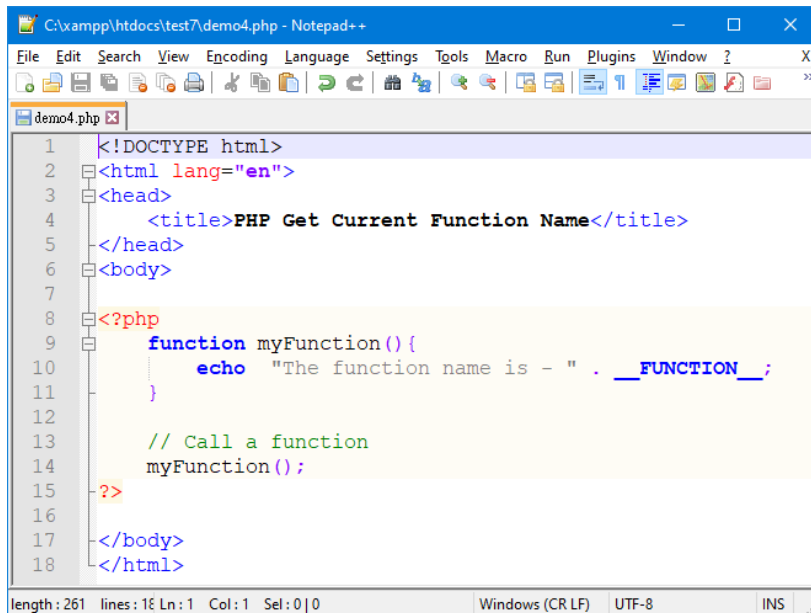
```
1 <?php
2 // Displays the directory of this file
3 echo "The directory of this file is: " . __DIR__;
4 ?>
```

length: 100 lines Ln: 4 Col: 3 Sel: 0 | 0 Windows (CR LF) UTF-8 INS

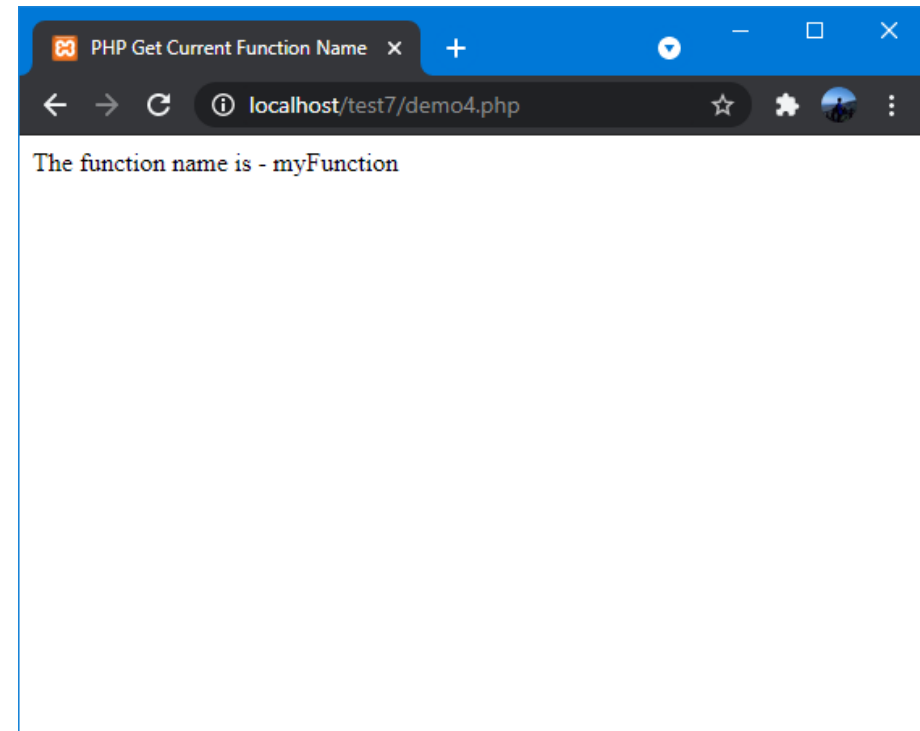


4. Konstanta `__FUNCTION__`

Konstanta `__FUNCTION__` akan berisi nama fungsi (*constant returns the name of the current function*).



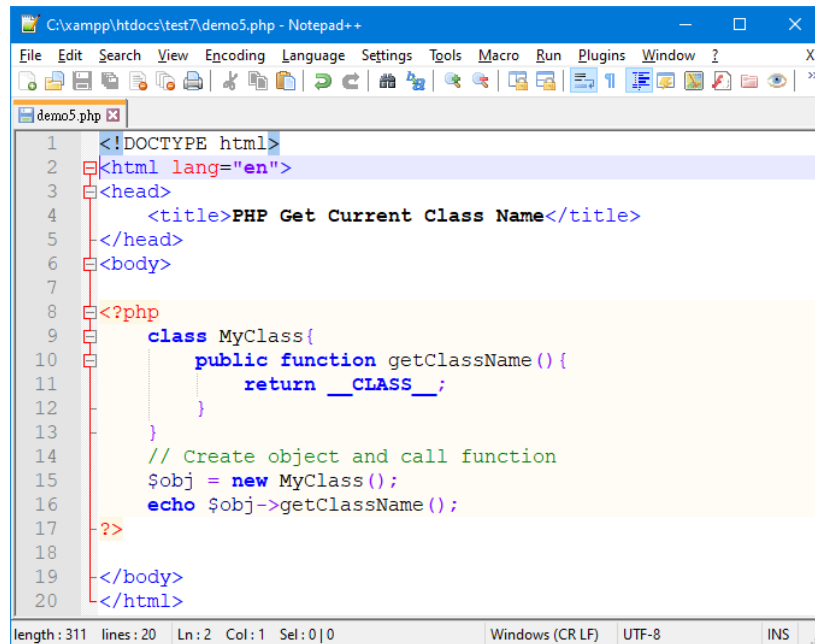
```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>PHP Get Current Function Name</title>
5 </head>
6 <body>
7
8 <?php
9   function myFunction(){
10     echo "The function name is - " . __FUNCTION__;
11   }
12
13   // Call a function
14   myFunction();
15 >?>
16
17 </body>
18 </html>
```



```
PHP Get Current Function Name
localhost/test7/demo4.php
The function name is - myFunction
```

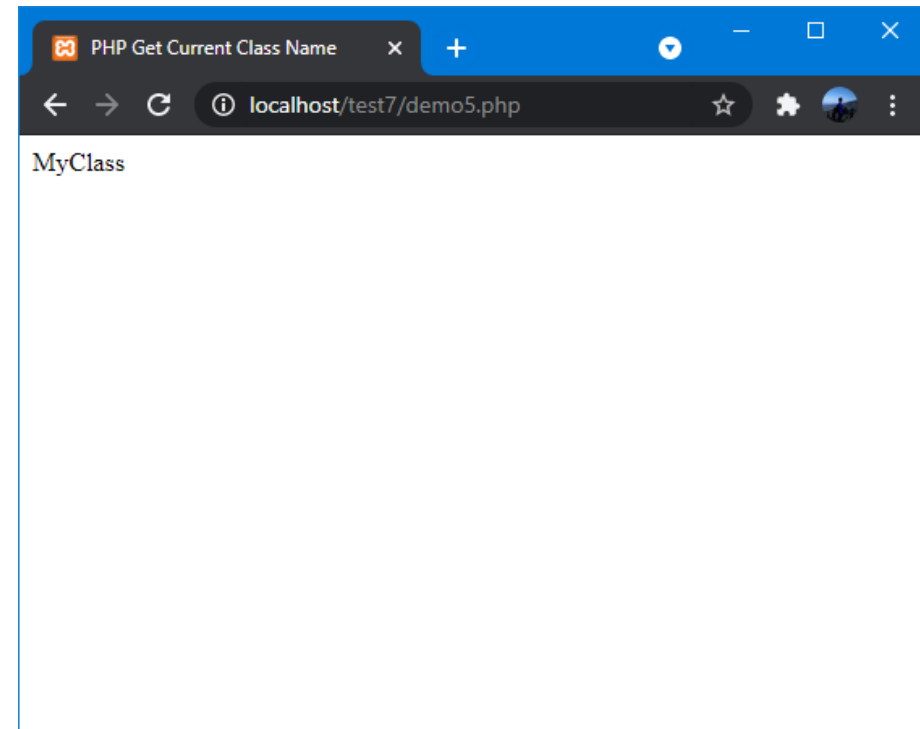
5. Konstanta `__CLASS__`

Konstanta `__CLASS__` akan berisi nama dari class (`__CLASS__` constant returns the name of the current class).



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <title>PHP Get Current Class Name</title>
5 </head>
6 <body>
7
8 <?php
9     class MyClass{
10         public function getClass_name(){
11             return __CLASS__;
12         }
13     }
14     // Create object and call function
15     $obj = new MyClass();
16     echo $obj->getClass_name();
17 -?>
18
19 </body>
20 </html>
```

length: 311 lines: 20 Ln: 2 Col: 1 Sel: 0|0 Windows (CR LF) UTF-8 INS



PHP Get Current Class Name

localhost/test7/demo5.php

MyClass

Trait in OOP PHP

Class

```
class ApaKabar
{
    public function apaKabar() {
        return "Apa kabar?";
    }
}

class Pesan extends ApaKabar
{
    //

}

$obj = new Pesan();
echo $obj->apaKabar();
```

Jika class **Pesan** ingin menggunakan method `apaKabar()` maka tinggal melakukan *extends* ke class `ApaKabar`.

Trait (reuse code)

```
trait ApaKabar
{
    public function apaKabar() {
        return "Apa kabar?";
    }
}

class Pesan
{
    // menggunakan use
    use ApaKabar;
}

$obj = new Pesan();
echo $obj->apaKabar();
```

Cara memakai *trait* yaitu dengan menggunakan keyword **use** dalam sebuah class dan diiringi nama trait.

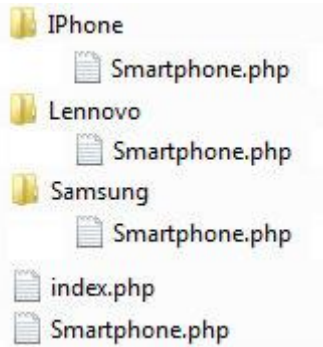
Namespace *in* OOP PHP

Namespace digunakan untuk menghindari *konflik* dari **penamaan Class yang sama**, karena terkadang para Developer menggunakan nama Class yang sama namun dalam kategori yang berbeda.

How to use Namespace in PHP ?

Link: <https://www.dumetschool.com/blog/Apa-itu-namespace-di-PHP>

Buatlah struktur folder dan file seperti berikut ini:



Edit file Smartphone.php

```
1 <?php
2 class Smartphone{
3
4     function PowerOn(){
5         echo "<p>Hello, I'm Smartphone</p>";
6     }
7
8 }
9
10 ?>
```

Edit file Smartphone.php

```
1 <?php
2 namespace iPhone;
3
4 class Smartphone{
5
6     function PowerOn(){
7         echo "<p>Hello, I'm iPhone</p>";
8     }
9
10 }
11
12 ?>
```

Edit file Smartphone.php

```
1 <?php
2 namespace Lenovo;
3
4 class Smartphone{
5
6     function PowerOn(){
7         echo "<p>Hello, I'm Lenovo</p>";
8     }
9
10 }
11
12 ?>
```

Edit file Smartphone.php

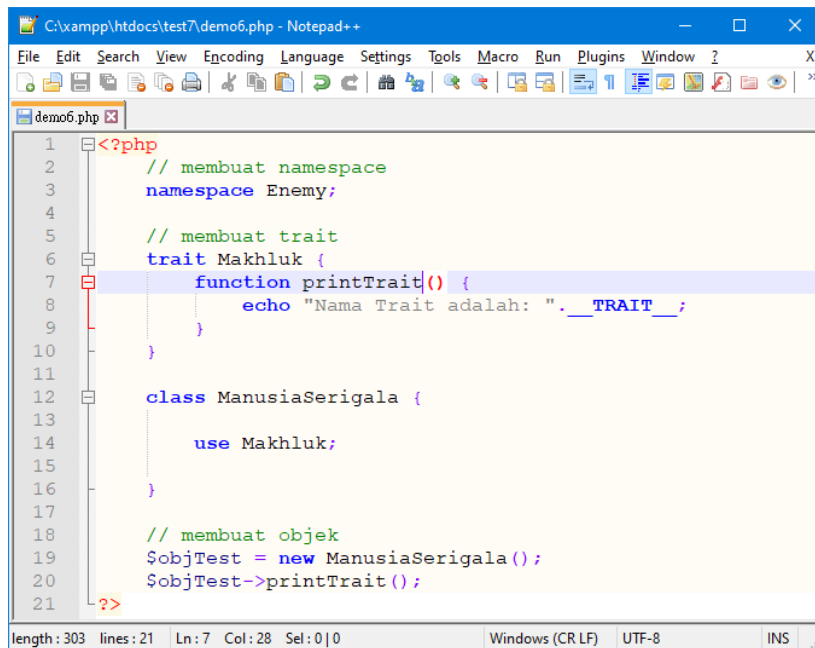
```
1 <?php
2 namespace Samsung;
3
4 class Smartphone{
5
6     function PowerOn(){
7         echo "<p>Hello, I'm Samsung</p>";
8     }
9
10 }
11
12 ?>
```

Terakhir, edit file index.php menjadi seperti berikut ini :

```
1 <?php
2 include_once 'Smartphone.php';
3 include_once 'iPhone/Smartphone.php';
4 include_once 'Samsung/Smartphone.php';
5 include_once 'Lenovo/Smartphone.php';
6
7 $Smartphone = new \Smartphone;
8 $Smartphone->PowerOn();
9
10 $Lenovo = new \Lenovo\Smartphone;
11 $Lenovo->PowerOn();
12
13 $Samsung = new \Samsung\Smartphone;
14 $Samsung->PowerOn();
15
16 $iPhone = new \iPhone\Smartphone;
17 $iPhone->PowerOn();
18 ?>
```

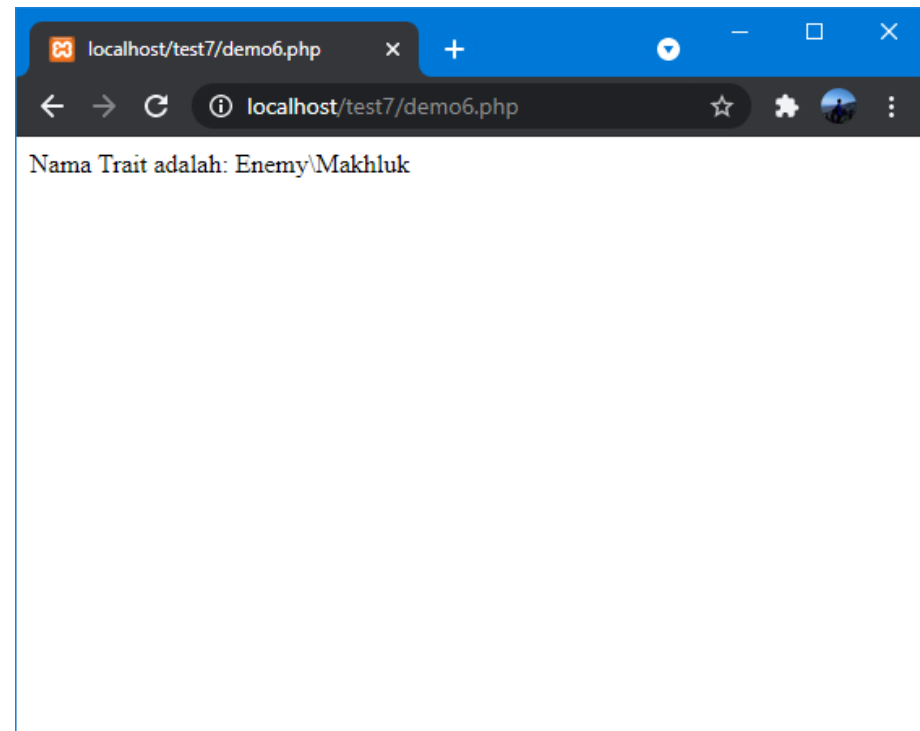
6. Konstanta `__TRAIT__`

Konstanta `__TRAIT__` akan berisi nama ***trait*** dan ***namespace***. ***Trait*** adalah suatu mekanisme dimana suatu class diizinkan untuk menggunakan kembali kode program (*code reuse*) yang berasal dari hirarki yang berbeda. Sedangkan, ***Namespace*** merupakan ***group*** atau kumpulan entities (*class, object, function*) yang dikelompokkan dalam satu nama. *Namespace* didalam PHP juga menggambarkan ***directory*** abstrak sebuah file.



```
1 <?php
2     // membuat namespace
3     namespace Enemy;
4
5     // membuat trait
6     trait Makhluk {
7         function printTrait() {
8             echo "Nama Trait adalah: " . __TRAIT__;
9         }
10    }
11
12    class ManusiaSerigala {
13
14        use Makhluk;
15
16    }
17
18    // membuat objek
19    $objTest = new ManusiaSerigala();
20    $objTest->printTrait();
21    ?>
```

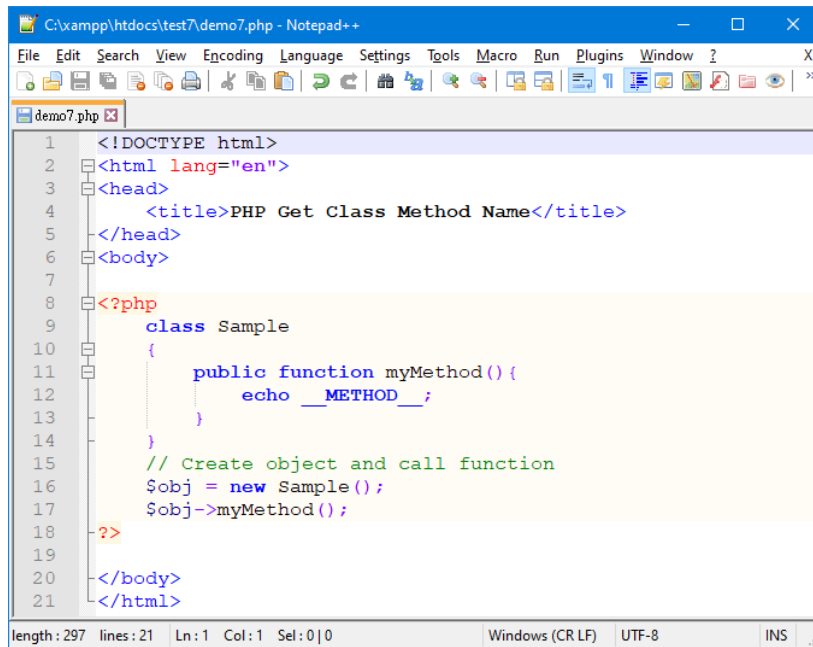
length: 303 lines: 21 Ln: 7 Col: 28 Sel: 0|0 Windows (CR LF) UTF-8 INS



```
localhost/test7/demo6.php
Nama Trait adalah: Enemy\Makhluk
```

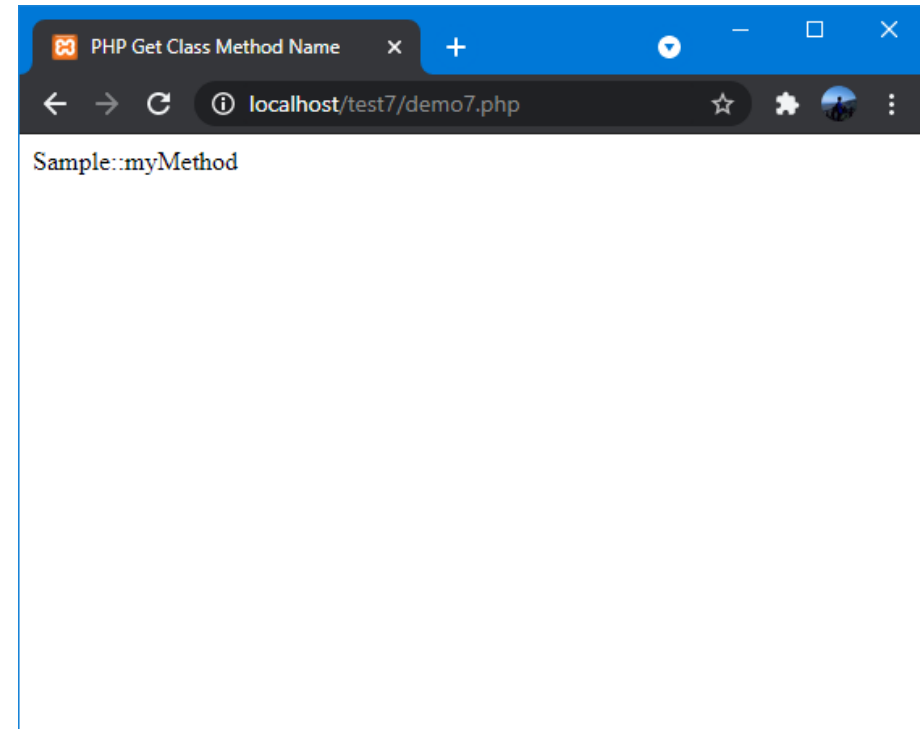
7. Konstanta `__METHOD__`

Konstanta `__METHOD__` sama seperti konstanta `__FUNCTION__`. Ia akan berisi nama *method* dan *class* pada *file* yang digunakan (*return the name of the current class method*).



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>PHP Get Class Method Name</title>
5 </head>
6 <body>
7
8 <?php
9   class Sample
10   {
11     public function myMethod() {
12       echo __METHOD__;
13     }
14   }
15   // Create object and call function
16   $obj = new Sample();
17   $obj->myMethod();
18 ?>
19
20 </body>
21 </html>
```

length: 297 lines: 21 Ln: 1 Col: 1 Sel: 0 | 0 Windows (CR LF) UTF-8 INS



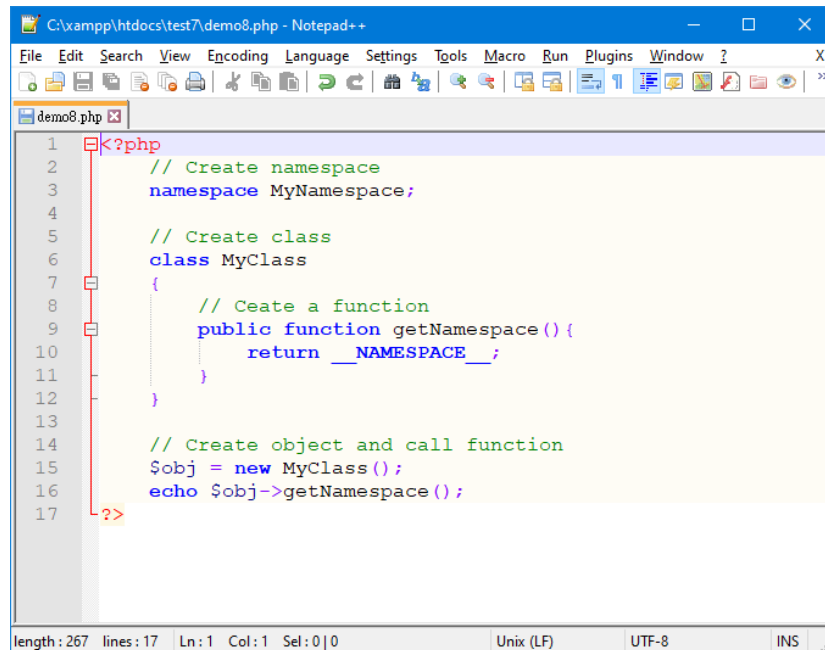
PHP Get Class Method Name

localhost/test7/demo7.php

Sample::myMethod

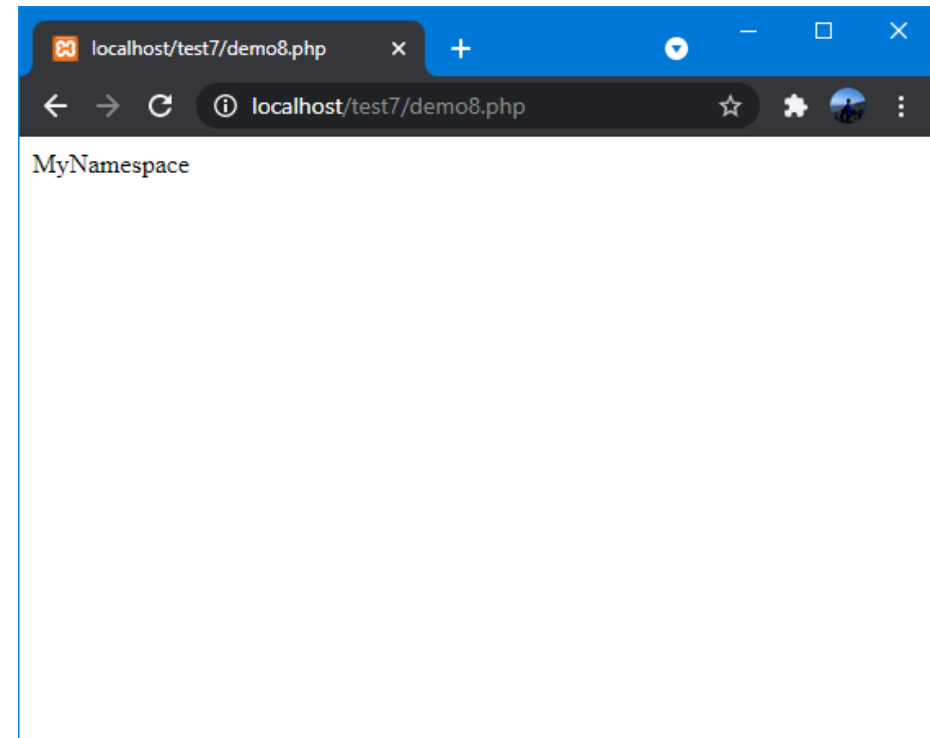
8. Konstanta `__NAMESPACE__`

Konstanta `__NAMESPACE__` akan berisi nama *namespace* (*return the name of the current namespace*).



```
1 <?php
2 // Create namespace
3 namespace MyNamespace;
4
5 // Create class
6 class MyClass
7 {
8     // Ceate a function
9     public function getNamespace(){
10         return __NAMESPACE__;
11     }
12 }
13
14 // Create object and call function
15 $obj = new MyClass();
16 echo $obj->getNamespace();
17 ?>
```

length: 267 lines: 17 Ln: 1 Col: 1 Sel: 0 | 0 Unix (LF) UTF-8 INS

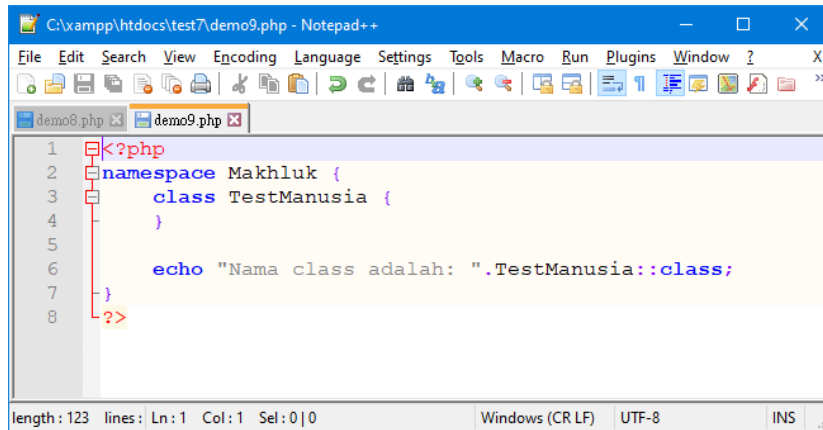


localhost/test7/demo8.php

MyNamespace

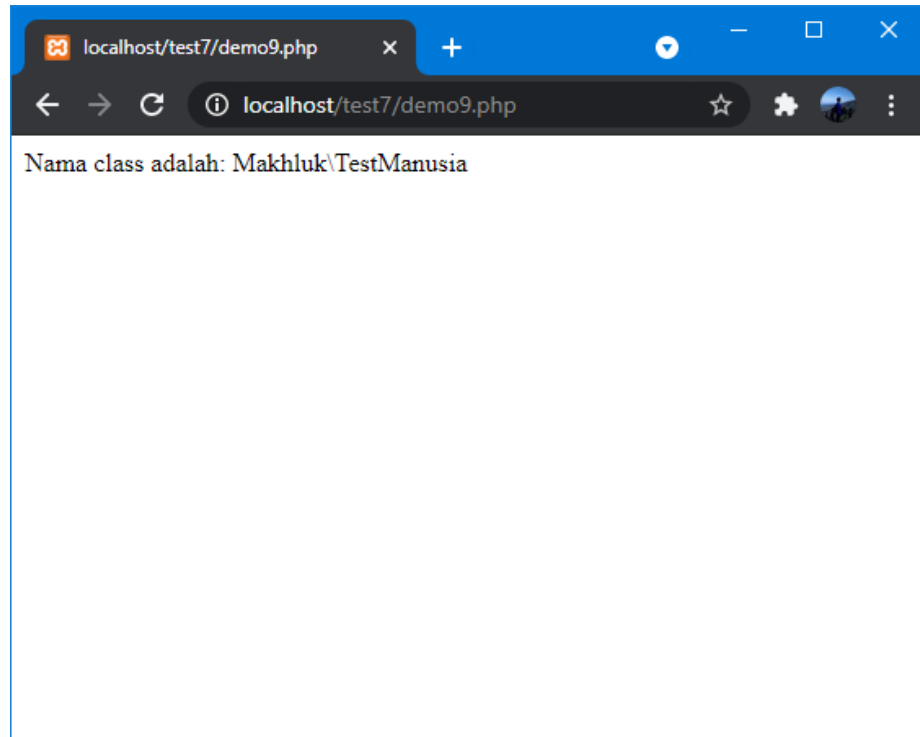
9. Konstanta ClassName::class

Hampir sama seperti konstanta `__TRAIT__`, konstanta `ClassName::class` akan berisi nama *class* dan *namespace*.



```
1 <?php
2 namespace Makhluk {
3     class TestManusia {
4     }
5
6     echo "Nama class adalah: ".TestManusia::class;
7 }
8 ?>
```

length: 123 lines: Ln: 1 Col: 1 Sel: 0 | 0 Windows (CR LF) UTF-8 INS



localhost/test7/demo9.php

Nama class adalah: Makhluk\TestManusia

Apakah sebuah constant di PHP dapat digunakan hanya untuk local saja? Seperti yang kita ketahui bersama bahwa constant biasanya bersifat global.





Exercise *for* Students

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture6\
Create file : latihan_.php

Buat contoh 1-5 Magic Constants di PHP yang sudah di pelajari sebelumnya.

References

PHP define() Function

https://www.w3schools.com/php/func_misc_define.asp

PHP Constants

https://www.w3schools.com/php/php_constants.asp

PHP | Magic Constants

<https://www.geeksforgeeks.org/php-magic-constants/>

Magic Constants

<https://www.javatpoint.com/php-magic-constants>

PHP - Constants Types

https://www.tutorialspoint.com/php/php_constants.htm

Apa itu Magic Constant ?

<https://www.babastudio.com/blog/Apa-itu-Magic-Constant>

PHP Magic Constants

<https://www.tutorialrepublic.com/php-tutorial/php-magic-constants.php>

Mengenai 9 Magic Constants di PHP

<https://www.petanikode.com/php-magic-constant/>

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

